2.1

p. 100 # 1*, 3, 5, 7, 9, 11, 18, 27

$A=\begin{bmatrix}2 & 0 & -1\\ 4 & -5 & 2\end{bmatrix}$ $B=\begin{bmatrix}7 & -5 & 1\\ 1 & -4 & -3\end{bmatrix}$

1. $-2A=\begin{bmatrix}-4 & 0 & 2\\ -8 & 10 & -4\end{bmatrix}$ $B-2A=\begin{bmatrix}3 & -5 & 3\\ -7 & 6 & -7\end{bmatrix}$

$AC=\begin{bmatrix}2 & 0 & -1\\ 4 & -5 & 2\end{bmatrix}\begin{bmatrix}1 & 2\\ -2 & 1\end{bmatrix}$ $(2\times3)(2\times2)$ Not possible

$CD=\begin{bmatrix}1 & 2\\ -2 & 1\end{bmatrix}\begin{bmatrix}3 & 5\\ -1 & 4\end{bmatrix}=\begin{bmatrix}1 & 13\\ -7 & -6\end{bmatrix}$

3. $A=\begin{bmatrix}2 & -5\\ 3 & -2\end{bmatrix}$ $3I_2=3\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$

$3I_2-A=\begin{bmatrix}1 & 5\\ -3 & 5\end{bmatrix}$ $3I_2A=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}\begin{bmatrix}2 & -5\\ 3 & -2\end{bmatrix}=\begin{bmatrix}6 & -15\\ 9 & -6\end{bmatrix}$

5. $Ab_1=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}\begin{bmatrix}4\\ -2\end{bmatrix}=\begin{bmatrix}-10\\ 0\\ 26\end{bmatrix}$ $Ab_2=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}\begin{bmatrix}-2\\ 3\end{bmatrix}=\begin{bmatrix}11\\ 8\\ -19\end{bmatrix}$

$AB=\begin{bmatrix}-1 & 3\\ 2 & 4\\ 5 & -3\end{bmatrix}\begin{bmatrix}4 & -2\\ -2 & 3\end{bmatrix}=\begin{bmatrix}-10 & 11\\ 0 & 8\\ 26 & -19\end{bmatrix}$

7. A: 5x3 AB: 5x7 $(5\times3)(3\times7)=(5\times7)$

9. $\begin{bmatrix}2 & 3\\ -1 & 1\end{bmatrix}\begin{bmatrix}1 & 9\\ -3 & k\end{bmatrix}=\begin{bmatrix}-7 & 18+3k\\ -4 & k-9\end{bmatrix}$

$18+3k=12$

$+3k=-6$

$k=2$

$-4=-6-k$

$2=-k$

$k=-2$

$\begin{bmatrix}1 & 9\\ -3 & k\end{bmatrix}\begin{bmatrix}2 & 3\\ -1 & 1\end{bmatrix}=\begin{bmatrix}-7 & 12\\ -6-k & k-9\end{bmatrix}$

$k=-2$

11. $AD = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}\begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 & 6 \\ 10 & 12 & 10 \\ 15 & 15 & 12 \end{bmatrix}$

$DA = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 5 & 10 & 15 \\ 6 & 12 & 15 \\ 6 & 10 & 12 \end{bmatrix}$

R mult does col.
L mult does row
so $AB = BA$ if $B = kI_3$ w/ $k \in \mathbb{R}$

18. $\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix}\begin{bmatrix} b_1 & b_2 & 0 \\ b_3 & b_4 & 0 \\ b_5 & b_6 & 0 \end{bmatrix} = \begin{bmatrix} ab & ab & 0 \\ ab & ab & 0 \\ ab & ab & 0 \end{bmatrix}$

all zeros

27. $u = \begin{bmatrix} -3 \\ 2 \\ -5 \end{bmatrix} \quad v = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$

$u^T v = \begin{bmatrix} -3 & 2 & -5 \end{bmatrix}\begin{bmatrix} a \\ b \\ c \end{bmatrix} = -3a + 2b - 5c \qquad v^T u = \begin{bmatrix} a & b & c \end{bmatrix}\begin{bmatrix} -3 \\ 2 \\ -5 \end{bmatrix}$

$= -3a + 2b - 5c$

$uv^T = \begin{bmatrix} -3 \\ 2 \\ -5 \end{bmatrix}\begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} -3a & -3b & -3c \\ 2a & 2b & 2c \\ -5a & -5b & -5c \end{bmatrix}$

$vu^T = \begin{bmatrix} a \\ b \\ c \end{bmatrix}\begin{bmatrix} -3 & 2 & -5 \end{bmatrix} = \begin{bmatrix} -3a & 2a & -5a \\ -3b & 2b & -5b \\ -3c & 2c & -5c \end{bmatrix}$

2.2

p. 109 # 1*, 5*, 7, 29, 31, 32

1. $\begin{bmatrix} 8 & 6 \\ 5 & 4 \end{bmatrix}^{-1} = \frac{1}{32-30}\begin{bmatrix} 4 & -6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}$

5. $\begin{bmatrix} 2 & -3 \\ -5/2 & 4 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ -9 \end{bmatrix}$

7. $A = \begin{bmatrix} 1 & 2 \\ 5 & 12 \end{bmatrix}$ $b_1 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$ $b_2 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$ $b_3 = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ $b_4 = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

a) $A^{-1} = \frac{1}{12-10}\begin{bmatrix} 12 & -2 \\ -5 & 1 \end{bmatrix} = \begin{bmatrix} 6 & -1 \\ -5/2 & 1/2 \end{bmatrix}$

$x_1 = \begin{bmatrix} -9 \\ 4 \end{bmatrix}$ $x_2 = \begin{bmatrix} 11 \\ -5 \end{bmatrix}$ $x_3 = \begin{bmatrix} 6 \\ -2 \end{bmatrix}$ $x_4 = \begin{bmatrix} 13 \\ -5 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 2 & -1 & 1 & 2 & 3 \\ 5 & 12 & 3 & -5 & 6 & 5 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 2 & 8 & -10 & -4 & -10 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & -1 & 1 & 2 & 3 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -9 & 11 & 6 & 13 \\ 0 & 1 & 4 & -5 & -2 & -5 \end{bmatrix}$

29. $\begin{bmatrix} 1 & -3 \\ 4 & -9 \end{bmatrix}^{-1} = \frac{1}{-9+12}\begin{bmatrix} -9 & 3 \\ -4 & 1 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -4/3 & 1/3 \end{bmatrix}$

31. $\begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ -3 & 1 & 4 & 0 & 1 & 0 \\ 2 & -3 & 4 & 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & -3 & 8 & -2 & 0 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 0 & -2 & 1 & 0 & 0 \\ 0 & 1 & -2 & 3 & 1 & 0 \\ 0 & 0 & 2 & 7 & 3 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 8 & 3 & 1 \\ 0 & 1 & 0 & 10 & 4 & 1 \\ 0 & 0 & 1 & 7/2 & 3/2 & 1/2 \end{bmatrix}$

32. $\begin{bmatrix} 1 & 2 & -1 & 1 & 0 & 0 \\ -4 & -7 & 3 & 0 & 1 & 0 \\ -2 & -6 & 4 & 0 & 0 & 1 \end{bmatrix}$

$\sim \begin{bmatrix} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 & 1 & 0 \\ 0 & -2 & 2 & 2 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & -1 & 4 & 1 & 0 \\ 0 & 0 & 0 & 10 & 2 & 1 \end{bmatrix}$

Not invertible

2.3

p. 115 # 1*, 5, 11, 15, 16, 17, 23

1. $\begin{bmatrix} 5 & 7 \\ -3 & -6 \end{bmatrix}$ det A = -30 + 21 yes

5. $\begin{bmatrix} 3 & 0 & -3 \\ 2 & 0 & 4 \\ -4 & 0 & 7 \end{bmatrix}$ No, not a pivot every column. lin. dep. col.

11. a) True
b) True
c) False (not all nxn)
d) True
e) True

15. No, 1 part IMT true, all true

16. A invert $\Rightarrow$ A^T invert $\Rightarrow$ IMT holds for A^T

17. No, then its lin. dep. so all IMT false

23. $Fx=0$ has a non trivial solution

2.4

p. 121 # 1, 3, 5, 10, 11, 21

1. $\begin{bmatrix} I & 0 \\ E & I \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & B \\ EA+C & EB+D \end{bmatrix}$

3. $\begin{bmatrix} 0 & I \\ I & 0 \end{bmatrix}\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} C & D \\ A & B \end{bmatrix}$

5. $\begin{bmatrix} A & B \\ C & 0 \end{bmatrix}\begin{bmatrix} I & 0 \\ X & Y \end{bmatrix} = \begin{bmatrix} A+BX & BY \\ C & 0 \end{bmatrix} = \begin{bmatrix} 0 & I \\ Z & 0 \end{bmatrix}$

$Z = C$

$BY = I$

$Y = B^{-1}$

$A + BX = 0$

$BX = -A$

$X = -B^{-1}A$

10. $\begin{bmatrix} I & 0 & 0 \\ A & I & 0 \\ B & D & I \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 & 0 \\ P & I & 0 \\ Q & R & I \end{bmatrix}$

$$\begin{bmatrix} I & 0 & 0 \\ A & I & 0 \\ B & D & I \end{bmatrix}\begin{bmatrix} I & 0 & 0 \\ P & I & 0 \\ Q & R & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ A+P & I & 0 \\ B+DP+Q & D+R & I \end{bmatrix} = \begin{bmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

$A + P = 0$

$P = -A$

$B + DP + Q = 0$

$Q = -B - DP$

$Q = -B - D(-A)$

$Q = -B + DA$

$D + R = 0$

$R = -D$

11. a) True

b) False, need to know partition sizes

21. a) $A^2 = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

b) $M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} A & 0 \\ I & -A \end{bmatrix}$

$A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}$

$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$

$C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$

$D = -A = \begin{bmatrix} -1 & 0 \\ -2 & 1 \end{bmatrix}$

$M^2 = \begin{bmatrix} A & 0 \\ I & -A \end{bmatrix}\begin{bmatrix} A & 0 \\ I & -A \end{bmatrix} = \begin{bmatrix} A^2 & 0 \\ A-A & A^2 \end{bmatrix} = \begin{bmatrix} I & 0 \\ 0 & I \end{bmatrix}$

$AX = b$

$LUx = b$

$Ly = b$

$Ux = y$

2.5

p. 129 #1, 5, 7, 11, 19

1. $[L\ b] = \begin{bmatrix} 1 & 0 & 0 & -7 \\ -1 & 1 & 0 & 5 \\ 2 & -5 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 6 \end{bmatrix}$ $\quad X = \begin{bmatrix} 3 \\ 4 \\ -6 \end{bmatrix}$

$[U\ y] = \begin{bmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & -1 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{bmatrix}$

5. $[L\ b] = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 3 & 1 & 0 & 0 & 6 \\ -1 & 0 & 1 & 0 & 0 \\ -3 & 4 & -2 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}$

$[u\ y] = \begin{bmatrix} 1 & -2 & -2 & -3 & 1 \\ 0 & -3 & 6 & 0 & 3 \\ 0 & 0 & 2 & 4 & 1 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 & 0 & 38 \\ 0 & 1 & 0 & 0 & 16 \\ 0 & 0 & 1 & 0 & 8.5 \\ 0 & 0 & 0 & 1 & -4 \end{bmatrix}$

7. $\begin{bmatrix} 2 & 5 \\ -3 & -4 \end{bmatrix} \overset{\frac{1}{2}R_1}{\sim} \begin{bmatrix} 1 & 5/2 \\ -3 & -4 \end{bmatrix} \overset{3R_1+R_2}{\sim} \begin{bmatrix} 1 & 5/2 \\ 0 & 7/2 \end{bmatrix} \overset{\substack{2R_1 \\ 2R_2}}{\sim} \begin{bmatrix} 2 & 5 \\ 0 & 7 \end{bmatrix} = U$

$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1/2 & 0 \\ 0 & 1/2 \end{bmatrix} \overset{-3R_1+R_2}{\sim} \begin{bmatrix} 1/2 & 0 \\ -3/2 & 1/2 \end{bmatrix} \overset{2R_1}{\sim} \begin{bmatrix} 1 & 0 \\ -3/2 & 1/2 \end{bmatrix} = L$

11. $\begin{bmatrix} 3 & 7 & 2 \\ 6 & 19 & 4 \\ -3 & -2 & 3 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & 2 \\ 0 & 5 & 0 \\ 0 & 5 & 5 \end{bmatrix} \sim \begin{bmatrix} 3 & 7 & 2 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix} = U$

$-2R_1+R_2=R_2$, $R_1+R_3=R_3$ $\qquad$ $-R_2+R_3=R_3$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix} = L$

$R_2+R_3=R_3$ $\qquad$ $2R_1+R_2=R_2$, $-R_1+R_3=R_3$

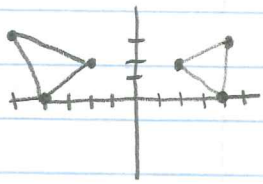
19. 0's not changed, only dealing w/ below pivots

$[A\ I]$ → I only changed on bottom as well so lower triangular

- By Theorem 7. in 2.2, this is A^{-1}

2.7 p. 144 #2, 3, 5, 7, 11, 19

2. $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 4 & 2 & 5 \\ 0 & 2 & 3 \end{bmatrix} = \begin{bmatrix} -4 & -2 & -5 \\ 0 & 2 & 3 \end{bmatrix}$



3. $\begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & -1 \\ 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

5. $\begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 & 0 \\ \sqrt{2}/2 & -\sqrt{2}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

7. $\begin{bmatrix} 1 & 0 & 6 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1/2 & -\sqrt{3}/2 & 0 \\ \sqrt{3}/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & -6 \\ 0 & 1 & -8 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1/2 & -\sqrt{3}/2 & 3+4\sqrt{3} \\ \sqrt{3}/2 & 1/2 & 4-3\sqrt{3} \\ 0 & 0 & 1 \end{bmatrix}$

11. $A_2A_1 = \begin{bmatrix} \sec\phi & -\tan\phi & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} \sec\phi - \tan\phi\sin\phi & -\tan\phi\cos\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} \cos\phi & -\sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$ which is $\mathbb{R}^2$ rotation matrix

19. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}\begin{bmatrix} 4.2 & 6 & 2 \\ 1.2 & 4 & 2 \\ 4 & 2 & 6 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 4.2 & 6 & 2 \\ 1.2 & 4 & 2 \\ 0 & 0 & 0 \\ .6 & .8 & .4 \end{bmatrix}$

$(4.2, 1.2, 0)/.6 \rightarrow (7, 2, 0)$
$(6, 4, 0)/.8 \rightarrow (7.5, 5, 0)$
$(2, 2, 0)/.4 \rightarrow (5, 5, 0)$